INSTITUTE	**FACULTY OF TECHNOLOGY**
PROGRAM	**BACHELOR OF TECHNOLOGY (MECHANICAL ENGINEERING)**
SEMESTER	**2**
COURSE TITLE	**COMPUTER PROGRAMMING**
COURSE CODE	**01CE1101**
COURSE CREDITS	**5**

Objective:

1. Students are expected to learn basics of Computer Programming which will help them to apply these concepts in day-to-day life. The course discusses various notations that are required for developing algorithm in C language, which is used in many commercial as well as industrial applications.
2. Students are expected to learn basics of Computer Programming which will help them to apply these concepts in day-to-day life. The course discusses various notations that required for developing algorithm and for C language, which is used in many commercial, industrial as well as industrial applications. Keeping in view wide applications of files, a special unit of files is introduced.
3. Students are expected to learn basics of Computer Programming which will help them to apply these concepts in day-to-day life. The course discusses various notations that required for developing algorithm and for C language, which is used in many commercial, industrial as well as industrial applications. Keeping in view wide applications of files, a special unit of files is introduced.

Course Outcomes: After completion of this course, student will be able to:

1. Able to explain programming problems logically through flow charts and algorithms.
2. Identify programming principles using C Language.
3. Demonstrate problem solving skills through C Language.
4. Generate computer-based solution for real time problem using programming language.
5. Develop confidence to self-educate new programming languages.

Pre-requisite of course:NA

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
4	0	2	50	30	20	25	25

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Contents : Unit	Topics	Contact Hours
1	**Introduction** File Management:, Languages Low level – high Number level, System – Binary – Decimal conversion problems, Flowchart, Algorithm, Problem solving using flowchart and algorithm	6
2	**C Programming Basics** Introduction to C Programming, Structure of 'C' program, Compilation and linking processes, Constants, Variables, Data Types, C Tokens, Expression using operators in 'C', Type Conversion and Type Casting	7
3	**Control Structure and Looping** Decision Making statements, Decision Making statements, Conditional operator, Looping – Entry and Exit control loops, Concept of jump, Break and continue	8
4	**Array and String** Declaration and initialization of array, Types of arrays, Sorting and matrix operation using array, Strings – string operations, String array, String functions	9
5	**Functions** Functions – Definition of function, Declaration of function, Call by value, Call by references, Recursion	8
6	**Structure and Union** Need of structure data type, Structure definition, structure declaration, Structure within structure, Difference between structure and union	4
7	**Pointers and Dynamic Memory Allocation** Pointers – Definition, Initialization, Pointer arithmetic, Pointer and array, Chain of pointer, DMA concepts, DMA functions, malloc(), calloc(), realloc(), free()	5
8	**File Management** Introduction to file management, File Handling function	3
Total Hours		**50**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Practical 1** a. Motivational videos on Programming b. Introduction to Turbo C++ IDE c. Develop a program to print "Hello World" d. Develop a program to print your address i) using single printf ii) using multiple printf e. Develop a program to print addition of 2 numbers (with scanf)	2

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
2	**Practical 2** a. Develop a program to convert temperature from Fahrenheit to Celsius (Formula: f = 1.8 * c + 32) \n b. Develop a program to find percentage of 5 subjects c. Develop a program to convert seconds into hours, minutes & seconds and print in HH:MM:SS [e.g. 10000 seconds mean 2:46:40 (2 Hours, 46 Minutes, 40 Seconds)]	2
3	**Practical 3** Using simple if a. Develop a program to check whether the given number is positive or negative b. Develop a program to find out largest number from given three numbers using Logical Operator (&&) Using if…else… a. Develop a program to check whether the given number is positive or negative Using nested if a. Develop a program to find out largest number from given three numbers without using Logical Operator (&&)	2
4	**Practical 4** Using if…else if… else a. Develop a program to check whether the given year is leap year or not. [If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year] b. Develop a program to perform Addition, Subtraction, Multiplication and Division of 2 numbers as per user's choice c. Three sides of a triangle are entered through the keyboard, develop a program to check whether the triangle is isosceles, equilateral, scalene or right-angled triangle	2
5	**Practical 5** Using Conditional operator (expr1?expr2:expr3) a. Develop a program to find out largest number from given 2 numbers b. Develop a program to read 3 numbers, multiply largest number from first two numbers to third one Using Switch statement a. Develop a program to print day name based on day number	2
6	**Practical 6** Discuss while loop a. Develop a program to print 1 to 10 b. Develop a program to find factorial of the given number c. Develop a program to find whether the given number is prime or not using break d. Develop a program to print digits of given number e. Develop a program to check whether the given number is palindrome or not f. Develop a program to check whether the given number is Armstrong or not	2
7	**Practical 7** Discuss for loop a. Develop a program to print numbers from 1 to 10 b. Develop a program to print odd numbers from 1 to n c. Develop a program to calculate the sum of first n natural numbers d. Develop a program to display the n terms of odd natural number and their sum e. Develop a program to find factors of the given number	2

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
8	**Practical 8** Discuss nested for loop a. Develop a program to find the sum of 1 + (1+2) + (1+2+3) + (1+2+3+4)+ …+(1+2+3+4+….+n) b. Develop a program to compute the value of ???? by using the formula ????=1+ ??1!+??22!+??33!+??44!+? c. Develop a program to find out prime numbers between given two numbers	2
9	**Practical 9** a. Develop a program to display following patterns 1 1 1 * * * * * * * * * * * * * * * 2 3 0 1 A B * * * * * * * * * * * * 4 5 6 1 0 1 2 3 4 * * * * * * * * * * * * * * * 7 8 9 10 0 1 0 1 C D E F * * * * * * * * * * * * * * * * b. Develop a program to draw Pascal's triangle	2
10	**Practical 10** a. Develop a program to count number of positive or negative number from an array of n numbers b. Develop a program to read n numbers in an array and print them in reverse order c. Develop a program to find Max, Min, Sum, Avg of given numbers from an array d. Develop a program to sort elements of an array in an ascending order	2
11	**Practical 11** a. Develop a program to read values in two-dimensional array and print them in matrix form b. Develop a program to count number of positive, negative and zero elements from 3 X 3 matrix c. Develop a program to perform addition of two matrices d. Develop a program to calculate the length of given string using strlen e. Develop a program to copy string using strcpy f. Develop a program to compare two strings using strcmp g. Develop a program to concatenate two strings using strcat h. Develop a program to reverse string using strrev	2
12	**Practical 12** a. Develop a program to count simple interest using function b. Develop a program to generate Fibonacci series of N given number using function name fibbo. (e.g. 0 1 1 2 3 5 8…) c. Develop a program to find the factorial of a given number using recursion	2
13	**Practical 13** Structure and Union a. Develop a program to create structure of book with book title, author name, publication, and price. Read data of n books and display them. b. Develop a program to access members of a union using union variable Pointers a. Develop a program to print value and address of a variable b. Develop a program to swap value of two numbers using pointer	2
14	**Practical 14** a. Develop a program to display content of a file b. Develop a program to copy source file to destination file	2
Total Hours		**28**

Textbook :

1 Programming in ANSI C, E. Balagurusamy, McGraw Hill Education (India) Private Limited, 2019

2 Programming with ANSI and Turbo C, Ashok Kamthane, Pearson Education, 2006

References:

1 Ansi C Programming, Ansi C Programming, Yashavant P. Kanetkar, BPB Publications, 2010

2 Programming with C, Programming with C, Byron Gottfired, Tata McGraw-Hill, 2010

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
20.00	20.00	30.00	15.00	10.00	5.00

Instructional Method:

1 The course delivery method will depend upon the requirement of content and need of students. The teacher in addition to conventional teaching method by black board, may also use any of tools such as demonstration, role play, Quiz, brainstorming, MOOCs etc.

2 The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.

3 Practical examination will be conducted at the end of semester for evaluation of performance of students in laboratory.

4 Students will use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory.

Supplementary Resources:

1 http://nptel.ac.in/courses/106104128/

2 http://nptel.ac.in/courses/106106133/

3 https://cse02-iiith.vlabs.ac.in/

4 https://www.learn-c.org/

5 https://www.learn-c.org/